The Theory of Subspaces

Video 10.1.1

Now we'll focus on developing a theory of *subspaces*.

What sort of things do we want to understand and do?

- How can we build examples of subspaces?
- Can we find *data* to describe a subspace that is given to us?
- How can we compare given subspaces and decide if they are the same or not?
- What properties can we "measure" about a subspace?
- The most important property of a subspace seems to be some idea of *dimension*. We want to say something like: "a line has dimension 1, a plane has dimension 2, etc…". What do we mean by *dimension*? Can we develop a logical way to define dimension for every subspace?

The theory of subspaces is built on two essential concepts which we will focus on in turn:

Essential concept #1:

The **span** of a set of vectors.

Essential concept #2:

Linear independence of a list of vectors.

We begin by developing the ideas we need for span. The starting point is the concept of a **linear combination of vectors.**

Linear combinations.

Definition ('A linear combination of vectors').

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

A vector $\vec{w}$ is said to be a linear combination of these vectors if there exist scalars

$$c_1, \dots, c_m \in \mathbb{R}$$

such that

$$\vec{w} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m.$$

The idea again, in words:
If we say a vector $\vec{w}$ is a linear combination of some vectors

$$\vec{v}_1, \dots, \vec{v}_m$$

then we are saying that "$\vec{w}$ can be built from the building blocks $\vec{v}_1, \dots, \vec{v}_m$."

Example

Consider the following list of two vectors in $\mathbb{R}^4$:

$$(1,0,1,0), (1,1,-1,-1) \in \mathbb{R}^4.$$

Here are a few examples of linear combinations of these vectors:

$$(2)(1,0,1,0) + (-5)(1,1,-1,-1) = (-3,-5,7,5)$$

$$(10)(1,0,1,0) + (1)(1,1,-1,-1) = (11,1,9,-1)$$

$$(0.5)(1,0,1,0) + (-0.1)(1,1,-1,-1) = (0.4,-0.1,0.6,0.1)$$

$$\vdots$$

We say that these vectors arise as **linear combinations** of $(1,0,1,0)$ and $(1,1,-1-1)$. Of course there are infinitely many vectors like this.

Example

We are considering the following list of two vectors in $\mathbb{R}^4$:

$$(1,0,1,0), (1,1,-1,-1) \in \mathbb{R}^4.$$

Note two important special cases of this construction.

Special case #1:
The vectors on the list themselves can arise as linear combinations of the vectors on the list:

$$(1)(1,0,1,0) + (0)(1,1,-1,-1) = (1,0,1,0)$$

$$(0)(1,0,1,0) + (1)(1,1,-1,-1) = (1,1,-1,-1)$$

Example

We are considering the following list of two vectors in $\mathbb{R}^4$:

$$(1,0,1,0), (1,1,-1,-1) \in \mathbb{R}^4.$$

Special case #2:

The zero vector $\vec{0}$ is always a linear combination of any given non-empty list of vectors:

$$(0)(1,0,1,0) + (0)(1,1,-1,-1) = (0,0,0,0)$$

A first computational problem:

Video 10.1.2

Something we'll need to be able to do is decide if a vector we are given is a linear combination of some given list of vectors or not.

Actually this doesn't need any new ideas: we just need to recognize that this problem is just another example of a system of linear equations to solve.

Example

Consider the following list of two vectors in $\mathbb{R}^4$ which we were just studying:

$$(1,0,1,0), (1,1,-1,-1) \in \mathbb{R}^4.$$

Is the vector $(1,-2,5,2)$ a linear combination of these two vectors?

Solution

The given vector will be a linear combination of the given two vectors precisely when there are constants c_1 and c_2 making the following equation true:

$$(1,-2,5,2) = c_1(1,0,1,0) + c_2(1,1,-1,-1) \quad (\star).$$

Do such constants exist? How can we decide?

The important thing is just to notice that this question is just an example of a system of linear equations. If that is true then of course we know how to solve such systems using Gaussian elimination or something like that.

Solution (cont.)

Consider the equation we are trying to solve

$$(1,-2,5,2) = c_1(1,0,1,0) + c_2(1,1,-1,-1) \quad (\star).$$

If we evaluate the RHS into a single vector we get:

$$(c_1 + c_2, c_2, c_1 - c_2, -c_2).$$

Now equate the RHS to the LHS. These two vectors are equal if and only if all their components are equal. So every component becomes a separate linear equation. The 1 vector equation is equivalent to the following system of 4 linear equations:

$$\begin{cases} c_1 + c_2 & = & 1 \\ c_2 & = & -2 \\ c_1 - c_2 & = & 5 \\ -c_2 & = & 2 \end{cases}.$$

How many solutions does this system have?

Solution (cont.)

The augmented matrix is: $\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 1 & -1 & 5 \\ 0 & -1 & 2 \end{array}\right]$.

Gaussian elimination gives: $\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$. → Row echelon form

This system is consistent so there are indeed solutions, and we already know the answer to our question is yes.
We don't actually need it, but note that there is exactly one solution: $c_2 = -2$ and $c_1 = 3$.

Substituting this solution back into $(\star)$ we can confirm that:

$$(1,-2,5,2) = (3)(1,0,1,0) + (-2)(1,1,-1,-1)$$

In other words: the vector $(1,-2,5,2)$ is indeed a linear combination of the given two vectors. $\square$

Let's try another example:

Video 10.1.3

Exercise

Consider the following list of three vectors in $\mathbb{R}^4$:

$$(1,-1,0,1), (-1,2,-1,0), (0,1,-1,1) \in \mathbb{R}^4.$$

Is the vector $(1,-2,5,2)$ a linear combination of these three vectors?

Solution

It is a linear combination precisely if there are scalars c, d, e such that

$$(1,-2,5,2)$$
$$= c(1,-1,0,1) + d(-1,2,-1,0) + e(0,1,-1,1) \quad (\star).$$

Solution (continued):

If we compute the right hand side of this equation we get:
$(c-d, -c+2d+e, -d-e, c+e)$.

Equating this vector to the LHS vector $(1,-2,5,2)$ gives a system of four linear equations obtained by equating corresponding entries

$$\begin{array}{rcl} c-d & = & 1 \\ -c+2d+e & = & -2 \\ -d-e & = & 5 \\ c+e & = & 2 \end{array}.$$

The corresponding augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & -2 \\ 0 & -1 & -1 & 5 \\ 1 & 0 & 1 & 2 \end{array}\right]$$

Solution (continued):

We do Gaussian elimination as usual to solve this system:

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ -1 & 2 & 1 & -2 \\ 0 & -1 & -1 & 5 \\ 1 & 0 & 1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 5 \\ 0 & 1 & 1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \text{inconsistent!}$$

Looking at the last non-zero row we see immediately that this system is inconsistent. (Recall that that row corresponds to the equation $0c+0d+0e=4$ which has no solutions.)

So there are ***no*** choices of c, d, e which make the equation $(\star)$ true:

$$(1,-2,5,2) = c(1,-1,0,1) + d(-1,2,-1,0) + e(0,1,-1,1) \quad (\star).$$

So we conclude:
the vector $(1,-2,5,2)$ is ***not*** a linear combination of the vectors $(1,-1,0,1)$, $(-1,2,-1,0)$, and $(0,1,-1,1)$.

$\square$

Video 10.1.4

So what is the purpose of introducing this concept?

The first reason this concept is important is that we can use it to build a subspace.

Theorem (Taking all possible linear combinations.)

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

Now define a set S to be the set of all possible linear combinations of the vectors $\vec{v}_1, \dots, \vec{v}_m$. This is obviously a subset of $\mathbb{R}^n$.

Then:
the set S is not just a subset but a *subspace* of $\mathbb{R}^n$.

Proof

We are told S is the set of all possible linear combinations of the vectors

$$\vec{v}_1, \dots, \vec{v}_m.$$

Subspace axiom 1: Why does S contain the zero vector?

Because, as we noted earlier, the zero vector is always a linear combination of any given list of vectors:

$$(0, \dots, 0) = (0)\vec{v}_1 + \cdots + (0)\vec{v}_m.$$

Subspace axiom 2: Why is S closed under vector addition?

What do we need to do?
We need to show that given *any* two vectors from S, call them $\vec{x}$ and $\vec{y}$, that their sum $\vec{x}+\vec{y}$ is *also* a vector in S.

Because $\vec{x} \in S$ we know that their exists scalars $c_i \in \mathbb{R}$ such that

$$\vec{x} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m.$$

Similarly, because $\vec{y} \in S$ we know that their exists scalars $d_i \in \mathbb{R}$ such that

$$\vec{y} = d_1\vec{v}_1 + \cdots + d_m\vec{v}_m.$$

Thus:

$$\begin{aligned}\vec{x}+\vec{y} &= (c_1\vec{v}_1 + \cdots + c_m\vec{v}_m) + (d_1\vec{v}_1 + \cdots + d_m\vec{v}_m) \\ &= (c_1+d_1)\vec{v}_1 + \cdots + (c_m+d_m)\vec{v}_m\end{aligned}$$

Thus $\vec{x}+\vec{y} \in S$.

Subspace axiom 3: Why is S closed under scalar multiplication?

What do we need to do?
We need to show that given any vector from S, call it $\vec{x} \in S$, and any scalar $r \in \mathbb{R}$, that the corresponding scalar multiple $r\vec{v}$ again lies in S.

Because $\vec{v} \in S$ we know that their exist scalars c_i such that

$$\vec{x} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m.$$

Thus:

$$\begin{aligned}r\vec{x} &= r(c_1\vec{v}_1 + \cdots + c_m\vec{v}_m) \\ &= (rc_1)\vec{v}_1 + \cdots + (rc_m)\vec{v}_m\end{aligned}$$

Thus $r\vec{x} \in S$.

$\square$

Video 10.1.5

The span of a set of vectors

We just saw that taking "all possible linear combinations" of a given list of vectors builds a subspace. Let's give this important subspace a name:

Definition ('Span').

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite set S of vectors in this $\mathbb{R}^n$

$$S = \{\vec{v}_1, \dots, \vec{v}_m\} \in \mathbb{R}^n.$$

The **span** of this set, denoted

$$span(S) \quad \text{or} \quad span\{\vec{v}_1, \dots, \vec{v}_m\}$$

is defined to be subspace of $\mathbb{R}^n$ consisting of all possible linear combinations of this set of vectors.

Example

Consider a single vector $(1,2) \in \mathbb{R}^2$.

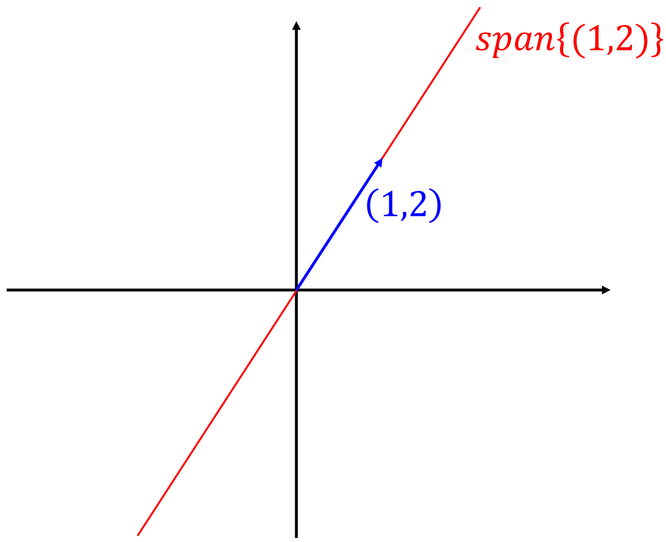
What is a "linear combination" of a set containing 1 vector?

Just a scalar multiple of that vector.

So the span of a set with just a single vector is the set of all possible scalar multiples of the vector.

Thus: $\mathrm{span}\{(1,2)\} = \{(\lambda, 2\lambda) \in \mathbb{R}^2; \lambda \in \mathbb{R}\}.$

Example (continued)



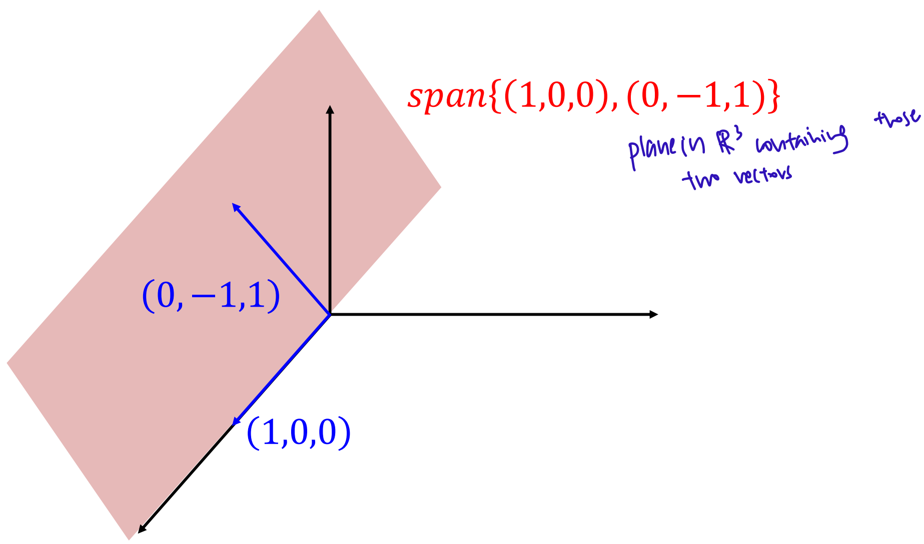
Example

Consider a pair of vectors $\{(1,0,0),(0,-1,1)\} \in \mathbb{R}^3$.

Then:

$$\mathrm{span}\{(1,0,0),(0,-1,1)\} = \{(\lambda,-\mu,\mu) \in \mathbb{R}^3; \lambda,\mu \in \mathbb{R}\}.$$

Example (continued)



So why is this idea "span" so important for us to develop a theory of subspaces?

Take a subspace (infinite) $V \subset \mathbb{R}^n$. $\Rightarrow$ Find a *finite* set S such that $V = span(S)$.

S is called a **spanning set** for V if this equation is true.

Comment:

Note that V is an infinite set and thus is impossible to work directly with. But S is a *finite amount of data* which completely determines the subspace V.

Instead of working directly with the infinite set V we can work with the finite list S. So for example we can give S to a computer when we want to compute something about V.

Exercise

Video 10.1.6

Consider the following system of linear equations:

$$\begin{cases} x_1 + 2x_2 + 3x_3 + 4x_4 & = & 0 \\ x_1 - x_3 - 2x_4 & = & 0 \end{cases}$$

Note that:

- This system is homogeneous.
- So the set of solutions V is a subspace of $\mathbb{R}^4$.

Find a finite set $S \subset \mathbb{R}^4$ such that $V = span(S)$.

Solution

A parameterization for the set of solutions will give us a spanning set automatically.

So let's solve this system. Gauss-Jordan elimination is a convenient way to write down a full parameterization for the set of solutions V:

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 1 & 0 & -1 & -2 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & -2 & -4 & -6 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 0 & 1 & 2 & 3 & 0 \end{array}\right].$$

So the reduced row echelon form of this system is:

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \end{array}\right]$$

This augmented matrix corresponds to the following system

$$\begin{cases} x_1 - x_3 - 2x_4 & = & 0 \\ x_2 + 2x_3 + 3x_4 & = & 0 \end{cases}.$$

Solution (continued):

The system is:

$$\begin{cases} x_1 - x_3 - 2x_4 &= 0 \\ x_2 + 2x_3 + 3x_4 &= 0 \end{cases}.$$

We can write down the solutions straight away in the way we have learnt. The variables x_3 and x_4 become free parameters $x_3 = s$ and $x_4 = t$ and we solve for the remaining variables in terms of them.

We deduce that the set of solutions V is given by:

$$\begin{cases} x_1 = s + 2t \\ x_2 = -2s - 3t \\ x_3 = s \\ x_4 = t \end{cases}, s, t \in \mathbb{R}.$$

To find a spanning set for V we just need to reinterpret this.

Solution (continued):

Rewrite this in set theory notation:

$$V = \{(s + 2t, -2s - 3t, s, t); s, t \in \mathbb{R}\}$$

Now rewrite the expression to "separate variables":

$$(s + 2t, -2s - 3t, s, t) = s(1, -2,1,0) + t(2, -3,0,1).$$

So we can express the set of solutions

$$V = \{\underbrace{s(1, -2,1,0) + t(2, -3,0,1); s, t \in \mathbb{R}}\}$$

Understand that this is set theory notation for the sentence "all possible linear combinations of the vectors $(1, -2,1,0)$ and $(2, -3,0,1)$".

Conclusion: A spanning set of the subspace of solutions of the given homogeneous system is

$$S = \{(1, -2,1,0), (2, -3,0,1)\}.$$

□

Video 10.1.7

How to compare spans?

We will use spanning sets to represent subspaces:

Take a subspace $V \subset \mathbb{R}^n$.	$\Rightarrow$	Find a *finite* set S such that $V = span(S)$.

span of S is V

There is a basic issue we need to understand:

We want to describe the infinite set V by a finite set S.

But it turns out a single subspace $V \subset \mathbb{R}^n$ can be described by many many many different spanning sets S.

For example we'll see later that the subspace $\mathbb{R}^2 \subset \mathbb{R}^2$ can be described in all of the following ways:

- $\mathbb{R}^2 = span\{(1,0),(0,1)\}$
- $\mathbb{R}^2 = span\{(1,1),(1,-1)\}$
- $\mathbb{R}^2 = span\{(1,0),(0,1),(1,1),(1,-1)\}$
- $\mathbb{R}^2 = span\{(1,0),(1,1),(1,2),(1,3),\dots,(1,10)\}$
- $\vdots$

How do we know when two subspaces are the same? How can we compare different spanning sets?

Video 10.1.8

How to compare spans?

For example:

We will see that the following two subspaces of $\mathbb{R}^2$ are equal:

S S'

$$span\{(1,0),(0,1)\} = span\{(1,1),(1,-1)\}$$

How could we tell these two subspaces are equal if all we have are their spanning sets

$S = \{(1,0),(0,1)\}$ and $S' = \{(1,1),(1,-1)\}$?

We will shortly meet a simple principle which will help us compare subspaces.

First, the simple main idea in words:

Recall that we should think of a spanning set S for a subspace V as a set of "building blocks" for V. Also let S' be a set of "building blocks" for another subspace V'.

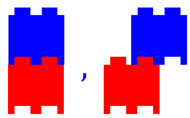
Question: If I tell you that every vector in S (the set of "building blocks" for V) is a linear combination of the vectors in S' (the set of building blocks for V') then what do you expect the relationship between V and V' to be?

Answer:

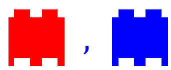
You should expect that $V \subset V'$.

Let's try a simple analogy with Lego:

- Let V denote the set of things I can build with the following set of Lego pieces:

  $S = \{$  $\}$

- Let V' denote the set of things I can build with the following set of Lego pieces:

  $S' = \{$  $\}$

Then for simple reasons we expect: $V \overset{\subset}{\boxed{?}} V'$.

> ***Theorem (Using spanning sets to compare subspaces.)***
>
> Fix some Euclidean space $\mathbb{R}^n$.
>
> Consider some finite set S of vectors in this $\mathbb{R}^n$ and let $V = span(S)$.
>
> Also let S' denote another finite set of vectors in this $\mathbb{R}^n$ and let $V' = span(S')$.
>
> Assume that $S \subset V'$.
>
> Then: $V \subset V'$.

In other words: This assumption says that every one of the spanning vectors for V is a linear combination of the spanning vectors for V'.

Proof

The proof is very simple. Like most proofs in this course, it is just an exercise in careful writing.

Let $S = \{\vec{v}_1, \ldots, \vec{v}_k\} \subset \mathbb{R}^n$.
And let $S' = \{\vec{w}_1, \ldots, \vec{w}_l\} \subset \mathbb{R}^n$.

What do we know?

We know that $S \subset span(S')$.

This says: for each $1 \leq i \leq k$, there exists constants $c_{i1}, \ldots, c_{il}$ such that

$$\vec{v}_i = c_{i1}\vec{w}_1 + \cdots + c_{il}\vec{w}_l \quad (\star).$$

Proof (continued).

What do we need to show? We want to show that $V \subset V'$.

The proof:

- Let $\vec{v}$ be an arbitrary element of V.
- Because $V = span(\vec{v}_1, \ldots, \vec{v}_k)$ we know that there exists scalars s_i such that
$$\vec{v} = s_1\vec{v}_1 + \cdots + s_k\vec{v}_k.$$
- Using $(\star)$ rewrite this:
$$\vec{v} = s_1(c_{i1}\vec{w}_1 + \cdots + c_{il}\vec{w}_l) + \cdots + s_k(c_{i1}\vec{w}_1 + \cdots + c_{il}\vec{w}_l)$$
- Now just rearrange this to group all the $\vec{w}_i$'s, using new symbols d_i for the coefficients that appear
$$\vec{v} = d_1\vec{w}_1 + \cdots + d_l\vec{w}_l.$$
- Finally, because $V' = span\{\vec{w}_1, \ldots, \vec{w}_l\}$ we observe that $v \in V'$.

$\square$

Let's go back and tackle this problem using this principle:

Example:

Show that the following two subspaces of $\mathbb{R}^2$ are in fact equal:

$$\underbrace{span\{(1,0),(0,1)\}}_{S} = \underbrace{span\{(1,1),(1,-1)\}}_{S'}$$

Solution:

We have to prove that two subsets of $\mathbb{R}^2$ are exactly equal to each other. (That is, they contain the same elements.)

There is a standard way to prove that subsets are equal in mathematics. We have to prove that each side is a subset of the other side.

Solution (continued):

So the proof naturally splits into two directions.

Step 1: Showing that

$$span\{(1,0),(0,1)\} \subset span\{(1,1),(1,-1)\}$$

* Are (1,0) and (0,1) linear combinations of (1,1) and (1,−1)?

Step 2: Showing that

$$span\{(1,0),(0,1)\} \supset span\{(1,1),(1,-1)\}$$

* Are (1,1) and (1,−1) linear combinations of (1,0) and (0,1)?

Proof of Step 1:

That $span\{(1,0),(0,1)\} \subset span\{(1,1),(1,-1)\}$

According to the previous theorem (which we called "Comparing subspaces using spanning sets") it is enough for us to prove that the spanning set of the left hand side (to be precise the two vectors $\{(1,0),(0,1)\}$) is contained in the full right hand side (to be precise all possible linear combinations of $\{(1,1),(1,-1)\}$.

So there are two facts to show:

a) $(1,0) \in span\{(1,1),(1,-1)\}$
b) $(0,1) \in span\{(1,1),(1,-1)\}$

Do there exist c_1 and c_2 such that
$(1,0) = c_1(1,1) + c_2(1,-1)$
$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -2 & -1 \end{array}\right]$ consistent row echelon form
This is consistent so $(1,0)$ can be constructed as a linear combination of $(1,1),(1,-1)$
$(1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$
Also $(0,1) = \frac{1}{2}(1,1) - \frac{1}{2}(1,-1)$

Proof of Step 1 (continued):

Part (a) of Step 1: Is $(1,0) \in span\{(1,1),(1,-1)\}$?

Let's answer this question using the method we learnt earlier. This vector $(1,0)$ lies in the span exactly when there exist scalars c_1 and c_2 such that

$$(1,0) = c_1(1,1) + c_2(1,-1).$$

Equating corresponding entries on the two sides gives us a system of two equations. The augmented matrix of this system is as follows. We can solve that matrix using Gaussian elimination: $\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & -2 & -1 \end{array}\right]$.

This system is in row echelon form, is consistent, and every column has a leading entry. Thus it definitely has solutions.

Thus we have confirmed that $(1,0) \in span\{(1,1),(1,-1)\}$.

Proof of Step 1 (continued):

Part (b) of Step 1: Is $(0,1) \in span\{(1,1),(1,-1)\}$?

The method is exactly analogous. In this case the linear combination equation gives the following system, which we solve as normal:

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & -1 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -2 & 1 \end{array}\right].$$

Again: This system is in row echelon form, is consistent, and every column has a leading entry. Thus it definitely has solutions.

Thus we have confirmed that $(0,1) \in span\{(1,1),(1,-1)\}$.

By Part (a) and Part (b), we conclude that indeed

$$span\{(1,0),(0,1)\} \subset span\{(1,1),(1,-1)\}.$$

Proof of Step 2:

That $span\{(1,0),(0,1)\} \supset span\{(1,1),(1,-1)\}$

We could solve this in the same way as we proved Step 1. Namely, according to the previous theorem ("Comparing subspaces using spanning sets") it is enough for us to prove the spanning set of the RHS is contained in the LHS. Namely:

a) $(1,1) \in span\{(1,0),(0,1)\}$
b) $(1,-1) \in span\{(1,0),(0,1)\}$

But in this case the spanning vectors on the RHS are very simple, so we can just prove these two facts directly.

Indeed the equation: $(1,1) = (1)(1,0) + (1)(0,1)$ immediately demonstrates that $(1,1) \in span\{(1,0),(0,1)\}$. → easy enough to see immediately.

Also, the equation: $(1,-1) = (1)(1,0) + (-1)(0,1)$ immediately demonstrates that $(1,1) \in span\{(1,0),(0,1)\}$.

Conclusion of the solution:

We have proved that:

1. $span\{(1,0),(0,1)\} \subset span\{(1,1),(1,-1)\}$
2. $span\{(1,0),(0,1)\} \supset span\{(1,1),(1,-1)\}$

Thus we conclude that:

$$span\{(1,0),(0,1)\} = span\{(1,1),(1,-1)\}.$$

$\square$

Summary of our solution to the previous problem:

These equations tell us
$span\{(1,0),(0,1)\} \subset span\{(1,1),(1,-1)\}$

$$(1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$$
$$(0,1) = \frac{1}{2}(1,1) - \frac{1}{2}(1,-1)$$

$$\left\{\begin{matrix}(1,0),\\(0,1)\end{matrix}\right\} \rightleftarrows \left\{\begin{matrix}(1,1),\\(1,-1)\end{matrix}\right\}$$

$$(1,1) = 1(1,0) + 1(0,1)$$
$$(1,-1) = 1(1,0) - 1(0,1)$$

These equations tell us
$span\{(1,0),(0,1)\} \cancel{\subset} span\{(1,1),(1,-1)\}$
$\supset$

Thus: $span\{(1,0),(0,1)\} = span\{(1,1),(1,-1)\}.$

Video 10.1.9

Matrix algorithms

Let's take these procedures we have been following and formulate them precisely as some matrix algorithms.

We'll develop and write down two closely related algorithms:

Algorithm #1: To decide if a given vector $\vec{w} \in \mathbb{R}^n$ lies in the span of a given set of vectors $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{R}^n$.

Algorithm #2: To decide if a given set of vectors $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{R}^n$ spans all of $\mathbb{R}^n$.

Algorithm #1.

The computational problem:

Does a given vector $\vec{w}$ lie in the span of a given set of vectors $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{R}^n$?

The input data:

- A positive integer n and a positive integer k.
- A list of k vectors in $\mathbb{R}^n$:

$$\begin{aligned} \vec{v}_1 &= (v_{11}, \dots, v_{1n}) \\ &\vdots \\ \vec{v}_k &= (v_{k1}, \dots, v_{kn}) \end{aligned}$$

v_{ij} → jth entry of the ith vector of the list

ith vector

jth component

- A vector $\vec{w} = (w_1, \dots, w_n) \in \mathbb{R}^n$.

Discussion:

If we try and solve this using the procedure we have been using so far, the first thing we do is write down the equation that expresses $\vec{w}$ as a linear combination.

Note that the variables we are trying to solve for here are c_1 through c_k. Everything else below is part of the input data:

$$(w_1, \dots, w_n) = c_1(v_{11}, \dots, v_{1n}) + \dots + c_k(v_{k1}, \dots, v_{kn}).$$

Solution exists

If there is a solution $(c_1, \dots, c_k)$ to this equation then **yes** the vector $\vec{w}$ does indeed lie in the span of the vectors $\vec{v}_1$ through $\vec{v}_k$.

No solution

If there is no solution $(c_1, \dots, c_k)$ to this equation then **no** the vector $\vec{w}$ does not lie in the span of the vectors $\vec{v}_1$ through $\vec{v}_k$.

Discussion (continued):

So. We need to try and solve this system for $(c_1, \dots, c_k)$:

$$(w_1, \dots, w_n) = c_1(v_{11}, \dots, v_{1n}) + \dots + c_k(v_{k1}, \dots, v_{kn}).$$

$$\left[\begin{array}{cccc|c} \uparrow & \uparrow & & \uparrow & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_k & \vec{w} \\ \downarrow & \downarrow & & \downarrow & \downarrow \end{array}\right]$$

Question:

Understand that this equation is equivalent to a system of n linear equations in the k variables $c_1, \dots, c_k$.

What is the augmented matrix corresponding to this linear system expressed in terms of the input data (the v_{ij}'s and w_k's) ?

Solution:

This is just a calculation, but anyway it is a good exercise in using indicial notation.

The given equation is:

$$(w_1, \dots, w_n) = c_1(v_{11}, \dots, v_{1n}) + \cdots + c_k(v_{k1}, \dots, v_{kn}).$$

Let's rewrite this using summation notation to make this precise: $(w_1, \dots, w_n) = \sum_{j=1}^{k} c_j(v_{j1}, \dots, v_{jn})$.

We can combine all the terms in the RHS into a single vector using the rules of vector operations:

$$\begin{aligned}(w_1, \dots, w_n) &= \sum_{j=1}^{k} (c_j v_{j1}, \dots, c_j v_{jn}) \\ &= \left(\sum_{j=1}^{k} c_j v_{j1}, \dots, \sum_{j=1}^{k} c_j v_{jn}\right)\end{aligned}$$

Solution (continued):

We just derived the equation:

$$(w_1, \dots, w_n) = \left(\sum_{j=1}^{k} c_j v_{j1}, \dots, \sum_{j=1}^{k} c_j v_{jn}\right).$$

This is equivalent to a system of n linear equations in the k variables $c_1, \dots, c_k$ which you get by equating corresponding positions on both sides.

$$\begin{cases} w_1 = \sum_{j=1}^{k} c_j v_{j1} \\ \quad \vdots \\ w_i = \sum_{j=1}^{k} c_j v_{ji} \qquad (\star). \\ \quad \vdots \\ w_n = \sum_{j=1}^{k} c_j v_{jn} \end{cases}$$

So what is the augmented matrix of this system?

Solution (continued):

Now we can just write the augmented matrix straight down from these equations in the usual way:

The entry in row i and column j of the augmented matrix is just the coefficient of c_j in equation i. Looking at $(\star)$ we see that that coefficient is just v_{ji}.

Thus the augmented matrix is: $\left[\begin{array}{ccc|c} v_{11} & \cdots & v_{k1} & w_1 \\ \vdots & \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} & w_n \end{array}\right]$.

Note that we can see the original vectors appearing as the ***columns*** of this matrix: $\left[\begin{array}{ccc|c} \uparrow & & \uparrow & \uparrow \\ \vec{v}_1 & \cdots & \vec{v}_k & \vec{w} \\ \downarrow & & \downarrow & \downarrow \end{array}\right]$.

vectors become columns in the matrix

vector you are investigating.

$\square$

Here is the algorithm:

Step 1

Construct an augmented matrix by making the vectors $\vec{v}_1 \ldots \vec{v}_k$ the columns of the matrix to the left of the partition, and making the vector $\vec{w}$ the column to the right of the partition:

$$\left[\begin{array}{ccc|c} \uparrow & & \uparrow & \uparrow \\ \vec{v}_1 & \cdots & \vec{v}_k & \vec{w} \\ \downarrow & & \downarrow & \downarrow \end{array}\right] = \left[\begin{array}{ccc|c} v_{11} & \cdots & v_{k1} & w_1 \\ \vdots & \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} & w_n \end{array}\right]$$

The algorithm (continued):

Step 2

Apply Gaussian elimination to put this matrix into row echelon form.

Step 3

There are two possibilities:

- If the row echelon form is consistent then **yes** the vector $\vec{w} \in span\{\vec{v}_1, \dots, \vec{v}_k\}$.
- If the row echelon form is not consistent then **no** the vector $\vec{w} \notin span\{\vec{v}_1, \dots, \vec{v}_k\}$.

Sometimes we want to tackle a number of vectors at once.

Video 10.1.10

Example:

Consider the following vectors in $\mathbb{R}^4$:

$$\vec{v}_1 = (1,-1,0,0)\,, \vec{v}_2 = (0,1,-1,0)\,, \vec{v}_3 = (0,0,1,-1).$$

Do any of the vectors

$$\vec{w}_1 = (2,1,-1,-2), \qquad \vec{w}_2 = (1,1,1,-1)$$

lie in $span\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$?

Solution:

The obvious thing to do would be to apply the algorithm twice.

But actually that would be repeating a lot of the operations.

Solution (continued):

A nice trick is just to put all the vectors $\vec{w}_i$ we are considering to the right of the partition. Then we just follow the algorithm as before. This performs the algorithm simultaneously for each vector. We can read off the results at the end.

In this example we write down:

$$\begin{array}{c} \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 2 & 1 \\ -1 & 1 & 0 & 1 & 1 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & -1 & -2 & -1 \end{array}\right] \\ \begin{array}{ccccc} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \vec{v}_1 & \vec{v}_2 & \vec{v}_3 & \vec{w}_1 & \vec{w}_2 \end{array} \end{array}$$

Solution (continued):

Then we do Gaussian elimination as usual:

$$\rightarrow \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & -1 & 1 & -1 & 1 \\ 0 & 0 & -1 & -2 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & -1 & -2 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|cc} 1 & 0 & 0 & 2 & 1 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right]$$

To answer for the specific vectors just focus on the relevant column.

Solution (continued):

The vector $\vec{w}_1$: Consider the matrix $\left[\begin{array}{ccc|c|c} 1 & 0 & 0 & 2 & \\ 0 & 1 & 0 & 3 & \text{[redacted]} \\ 0 & 0 & 1 & 2 & \\ 0 & 0 & 0 & 0 & \end{array}\right]$.

According to the algorithm: because this row echelon form is consistent we know that indeed the vector $\vec{w}_1$ is a linear combination of the vectors $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$.

The vector $\vec{w}_2$: Consider the matrix $\left[\begin{array}{ccc|c|c} 1 & 0 & 0 & & 1 \\ 0 & 1 & 0 & \text{[redacted]} & 2 \\ 0 & 0 & 1 & & 3 \\ 0 & 0 & 0 & & 2 \end{array}\right]$.

According to the algorithm: because this row echelon form is *inconsistent* we know that the vector $\vec{w}_2$ is NOT a linear combination of the vectors $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. $\square$

Algorithm #2.

Video 10.1.11

The previous algorithm decides whether a specific vector $\vec{w} \in \mathbb{R}^n$ lies in the span of some set $\{\vec{v}_1, \dots, \vec{v}_k\}$.

Now we'll use that algorithm to solve a related problem: Does, not just a specific vector, but *every* vector in $\mathbb{R}^n$ lie in the span of some given $\{\vec{v}_1, \dots, \vec{v}_k\}$.

The computational problem:

Does a given set of vectors $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{R}^n$ span the entire $\mathbb{R}^n$?

The input data:

- A positive integer n and a positive integer k.
- A list of k vectors in $\mathbb{R}^n$:

$$\begin{aligned} \vec{v}_1 &= (v_{11}, \dots, v_{1n}) \\ &\vdots \\ \vec{v}_k &= (v_{k1}, \dots, v_{kn}) \end{aligned}$$

First let's solve a specific example of this.
The example will show us how to formulate a matrix algorithm to solve this problem.

Example:

Consider the following set of three vectors in $\mathbb{R}^3$:

$$\{(1,2,3), (2,-1,3), (-1,3,0)\}.$$

Does this set span all of $\mathbb{R}^3$?

We need an expression that can stand in for any vector in $\mathbb{R}^3$.

Solution:

We need to decide whether *no matter what* vector $\vec{w} \in \mathbb{R}^3$ we choose that $\vec{w} \in span\{(1,2,3), (2,-1,3), (-1,3,0)\}$.

So we have to let $\vec{w}$ be a formula that can specialise to any possible vector in $\mathbb{R}^3$: $\vec{w} \in (w_1, w_2, w_3)$.

Then we run the algorithm from before, and see whether the answer depends on the choices for w_1, w_2, w_3.

Solution (continued).

Following the algorithm from before we write down the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & w_1 \\ 2 & -1 & 3 & w_2 \\ 3 & 3 & 0 & w_3 \end{array}\right].$$

These entries are undetermined constants because this column will represent every possible vector from $\mathbb{R}^3$.

Following the algorithm from before we apply Gaussian elimination to this augmented matrix:

$$\to \left[\begin{array}{ccc|c} 1 & 2 & -1 & w_1 \\ 0 & -5 & 5 & w_2 - 2w_1 \\ 0 & -3 & 3 & w_3 - 3w_1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & -1 & w_1 \\ 0 & -5 & 5 & w_2 - 2w_1 \\ 0 & 0 & 0 & w_3 - 3w_1 - \frac{3}{5}(w_2 - 2w_1) \end{array}\right]$$

Solution (continued).

So the row echelon form we end up with in this case is:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & w_1 \\ 0 & -5 & 5 & w_2 - 2w_1 \\ 0 & 0 & 0 & -\frac{9}{5}w_1 - \frac{3}{5}w_2 + w_3 \end{array}\right]$$

Recall how the algorithm works here. There are two possibilities.

1. If this matrix is consistent then indeed $\vec{w}$ lies in the span of the given vectors.
2. On the other hand: if the matrix is inconsistent, then $\vec{w}$ doesn't lie in the given span.

So for which $\vec{w}$ is this matrix consistent or inconsistent?

Solution (continued).

So for which $\vec{w}$ is this matrix consistent or inconsistent?

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & w_1 \\ 0 & -5 & 5 & w_2 - 2w_1 \\ 0 & 0 & 0 & -9/5w_1 - 3/5w_2 + w_3 \end{array}\right]$$

→ whether consistent or not depends on this term.

Case 1: $-\frac{9}{5}w_1 - \frac{3}{5}w_2 + w_3 = 0$
The system is consistent.
So $\vec{w}$ lies in the span.

Case 2: $-\frac{9}{5}w_1 - \frac{3}{5}w_2 + w_3 \neq 0$
The system is inconsistent.
So $\vec{w}$ does not lie in the span.

Now note that obviously sometimes $-\frac{9}{5}w_1 - \frac{3}{5}w_2 + w_3 \neq 0$. For example the case $\vec{w} = (1,0,0)$. In such cases such a $\vec{w}$ does not lie in the span.

So not every vector lies in the span. So we can now answer the problem:

$$span\{(1,2,3), (2,-1,3), (-1,3,0)\} \neq \mathbb{R}^3. \qquad \square$$

Discussion: How to formulate as a matrix algorithm?

Video 10.1.12

So let's now think about how to formulate this procedure as a clean matrix algorithm that we could give to a computer. Assume the given list of vectors $\{v_1, \dots, v_k\} \subset \mathbb{R}^n$ is:

$$\begin{array}{rcl} \vec{v}_1 & = & (v_{11}, \dots, v_{1n}) \\ & \vdots & \\ \vec{v}_k & = & (v_{k1}, \dots, v_{kn}) \end{array}$$

We let $\vec{w} = (w_1, \dots, w_n)$ denote an arbitrary vector in $\mathbb{R}^n$.

Following the algorithm we write down the following augmented matrix:

$$\left[\begin{array}{ccc|c} v_{11} & \cdots & v_{k1} & w_1 \\ \vdots & \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} & w_n \end{array}\right]$$

After Gaussian elimination there are two possibilities for the resulting row echelon form:

Case 1: There is a row which is all zero to the left of the partition.

Case 2: There is no such row.

Discussion of Case 1:

In this case the matrix will look like:

$$\left[\begin{array}{ccc|c} * & * & * & * \\ * & * & * & * \\ 0 & \cdots & 0 & w_i + (\text{linear function of other } w_j's) \end{array}\right].$$

Here i is the row index of the row where this row of zeros came from. (Be careful: Maybe it got swapped down from higher in the matrix during the Gaussian elimination procedure.)

Note that for some particular $\vec{w}$ (such as $w_i = 1$, all other $w_j = 0$) the expression in the bottom entry to the right of the partition is definitely $\neq 0$.

So in this case: the answer is NO, $span\{\vec{v}_1, \dots, \vec{v}_k\} \neq \mathbb{R}^n$.

After Gaussian elimination there are two possibilities for the resulting row echelon form:

Case 1: There is a row which is all zero to the left of the partition.

Case 2: There is no such row.

Discussion of Case 2:

In this case the system is obviously consistent no matter what values are chosen for the w_i's.

So in this case the answer is obviously YES

$$span\{\vec{v}_1, \dots, \vec{v}_k\} = \mathbb{R}^n.$$

We are ready to formulate the general matrix algorithm to decide if $span\{\vec{v}_1, \dots, \vec{v}_k\} = \mathbb{R}^n$.

The algorithm:

Step 1

Construct a matrix by making the vectors $\vec{v}_1 \dots \vec{v}_k$ the columns of the matrix.

$$\begin{bmatrix} \uparrow & & \uparrow \\ \vec{v}_1 & \cdots & \vec{v}_k \\ \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} v_{11} & \cdots & v_{k1} \\ \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} \end{bmatrix}$$

The algorithm (continued):

Step 2

Do Gaussian elimination to put this matrix into row echelon form M.

Step 3

There are two possibilities:

Case 1: The matrix M has at least one row consisting entirely of zeros. In this case the answer is **NO**:

$$span\{\vec{v}_1, \dots, \vec{v}_k\} \neq \mathbb{R}^n.$$

Case 2: The matrix M has no such all zero rows. In this case the answer is **YES**:

$$span\{\vec{v}_1, \dots, \vec{v}_k\} = \mathbb{R}^n.$$

□

Problem

Video 10.1.13

Just say we have k vectors $\{\vec{v}_1, \dots, \vec{v}_k\}$ in some $\mathbb{R}^n$.

Assume that $k < n$.

Is it possible that

$$span\{\vec{v}_1, \dots, \vec{v}_k\} = \mathbb{R}^n \ ?$$

Solution

Our intuition says NO. For example, we don't expect that we should be able to build all of $\mathbb{R}^3$ by taking combinations of just two vectors. But how to prove?

The algorithm tells us to write down the following matrix:

$$\underbrace{\begin{bmatrix} v_{11} & \cdots & v_{k1} \\ \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} \end{bmatrix}}_{k} \left.\vphantom{\begin{bmatrix} v_{11} \\ \vdots \\ v_{1n} \end{bmatrix}}\right\} n$$

Note that because $n > k$ this matrix is taller than it is wide.

Solution (continued)

Next the algorithm says to do Gaussian elimination.
The result of Gaussian elimination will be a matrix in row echelon form.

In a matrix in row echelon form:

$$(\#\, of\ non-zero\ rows) \leq (\#\, of\ columns).$$

(Why? Because for every row there is a different column containing its leading entry.)

So, because in this case we are told that $k < n$ then:

$$\underbrace{(\#\, of\ non-zero\ rows)}_{\text{in row echelon form}} \leq \underbrace{(\#\, of\ columns)}_{k} < \underbrace{(\#\, of\ rows)}_{n}.$$

Thus the row echelon form of the matrix of vectors definitely has zero rows.

According to the algorithm this means that the given set of vectors *cannot* span $\mathbb{R}^n$.

$\square$